

SOLAR CELLS

Molecular press annealing enables robust perovskite solar cells

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Thermal annealing improves the crystallinity of perovskite films and boosts their power conversion efficiencies (PCEs) in solar cells but also induces surface iodine loss and local lattice degradation. We demonstrate a molecular press annealing (MPA) strategy in which a 2-pyridylethylamine film is thermally and pressure-bonded to the perovskite surface. Real-time healing of iodine vacancies occurred during annealing and the lead-iodine framework was stabilized through optimized ligand engineering, resulting in enhanced structural integrity and long-term stability of perovskite films. This strategy enabled n-i-p perovskite solar cells to achieve a PCE of 26.6% (certified 26.5%). Notably, the devices retain 98.6 and 97.2% of their initial PCEs after 1617 hours of continuous operation under maximum power point tracking [ISOS-L-3 protocol, 85°C, 60% relative humidity (RH)] and 5280 hours of ambient storage (ISOS-D-1 protocol, room temperature, 10% RH).

Temperature governs perovskite crystallization by modulating nucleation and crystal growth dynamics. Thermal annealing enhances the crystallinity of polycrystalline perovskite thin films but also leads to the formation of undesired surface iodine vacancies (V_I) and PbI_2 nanocrystals (1–3). The soft ionic character of inorganic-organic leads to relatively weak lead-iodine (Pb–I) bonds (4–6) that can dissociate under photoexcitation (photon energies >2.5 eV) or thermal stress because of their low bond dissociation energy (142 kJ mol^{-1}) (7, 8). Consequently, Schottky-type defects involving V_I readily form in perovskites, especially at fragile surfaces during thermal annealing, and are accompanied by deep-level states originating from undercoordinated lead(II) ions and iodine-related species, including volatile iodine (I_2) (9, 10).

Excessive iodine loss can induce stress-driven collapse of the $[\text{PbI}_6]^{4-}$ octahedral framework that leads to perovskite degradation and a progressive loss of structural integrity (11, 12). In addition, shallow-level defects involving V_I accelerate ion migration, resulting in unfavorable n-type self-doping (13, 14). Moreover, although surface PbI_2 is widely recognized as a degradation byproduct during perovskite thermal annealing, its degradation dynamics remain poorly visualized, hindering a mechanistic understanding of perovskite instability.

Notably, surface reconstruction and lattice disorder increase surface defect concentrations by nearly two orders of magnitude compared with the bulk (15). To address these issues, various surface passivation strategies have been developed to mitigate perovskite defects (16). However, these approaches function as “post remediation” measures and fail to prevent degradation at its origin. Moreover, the use of polar solvents in solution-based passivation can disrupt the fragile perovskite

layer beneath, further compromising film stability (17). These limitations highlight the need for solvent-free strategies that can intrinsically suppress thermal decomposition during annealing-induced crystallization.

We studied the degradation dynamics of perovskites during annealing and show that V_I serve as initiators that trigger their decomposition into PbI_2 . This degradation proceeds as an intrinsic, dynamic process that inevitably accompanies thermal crystallization. To address this challenge, we developed a solid-state molecular press annealing (MPA) strategy that directly suppresses lattice degradation at its source. A compact layer of pyridine-based molecules was imprinted onto the perovskite top surface during annealing, providing sustained molecular-level confinement that preserved surface order and inhibited premature defect formation. 2-Pyridylethylamine (2-Pyy), an optimally tailored ligand, formed robust bidentate coordination with adjacent undercoordinated lead(II) ions throughout annealing, thereby stabilizing the Pb–I framework and suppressing the propagation of V_I defects. This coordination interaction preserved structural integrity, suppressed lattice collapse, and mitigated defect-induced ion migration to enhance device stability.

The MPA strategy enhanced charge collection and overall device performance in n-i-p perovskite solar cells (PSCs), achieving a power conversion efficiency (PCE) of 26.6% (certified: 26.5%) for a 0.08 cm^2 device and 24.9% for a 1 cm^2 device. Moreover, the devices demonstrated notable long-term stability, retaining 98.6% of their initial PCE after 1617 hours of continuous operation under maximum power point (MPP) tracking following the ISOS-L-3 protocol of 85°C, 60% relative humidity (RH), and 97.2% after 5280 hours of ambient storage under the ISOS-D-1 protocol (room temperature, 10% RH). In addition, the MPA strategy demonstrates broad universality across diverse perovskite systems and exhibits high reusability, enabling tens of imprinting cycles from a single pyridine-based template, reducing costs by more than 47 times compared with traditional solution-processed posttreatment.

The impacts of MPA on perovskite degradation kinetics

Previous studies have shown that PbI_2 nanocrystals inevitably form on perovskite surfaces during annealing-assisted crystallization, even for stoichiometric precursors ratios [such as FAI: PbI_2 = 1:1 molar ratio (FA, formamidinium)] or single-crystal materials (such as FAPbI_3 powders) (table S1). To overcome this challenge, we introduced the MPA strategy to regulate crystallization dynamics to mitigate PbI_2 formation and suppress crystal degradation.

The MPA treatment consists of two primary steps: A 2-Pyy film was first deposited on an indium tin oxide (ITO) substrate via a solution process (Fig. 1A). The prepared 2-Pyy imprinting film exhibits a high uniformity (fig. S1), enabling conformal contact with the perovskite film during MPA. After preannealing the perovskite film at 150°C for ~3 min, this 2-Pyy film was aligned in-plane and laminated face-to-face onto the perovskite surface (18). Following the annealing step, the imprinting film was peeled from one edge at a low angle in a single continuous motion for reuse (Fig. 1B). The MPA process, including temperature and pressure control, was managed using a hot press apparatus. Further experimental details are provided in the methods, fig. S2, and movie S1. Notably, based on the optimized MPA conditions (0.4 MPa, 100°C), the resulting perovskite films exhibit enhanced photoluminescence properties (fig. S3) and maintain clean detachment from the imprinting film (fig. S4).

To investigate the role of MPA in the annealing-induced degradation mechanism, we selected $\alpha\text{-FAPbI}_3$ (from stoichiometric FAI and PbI_2)

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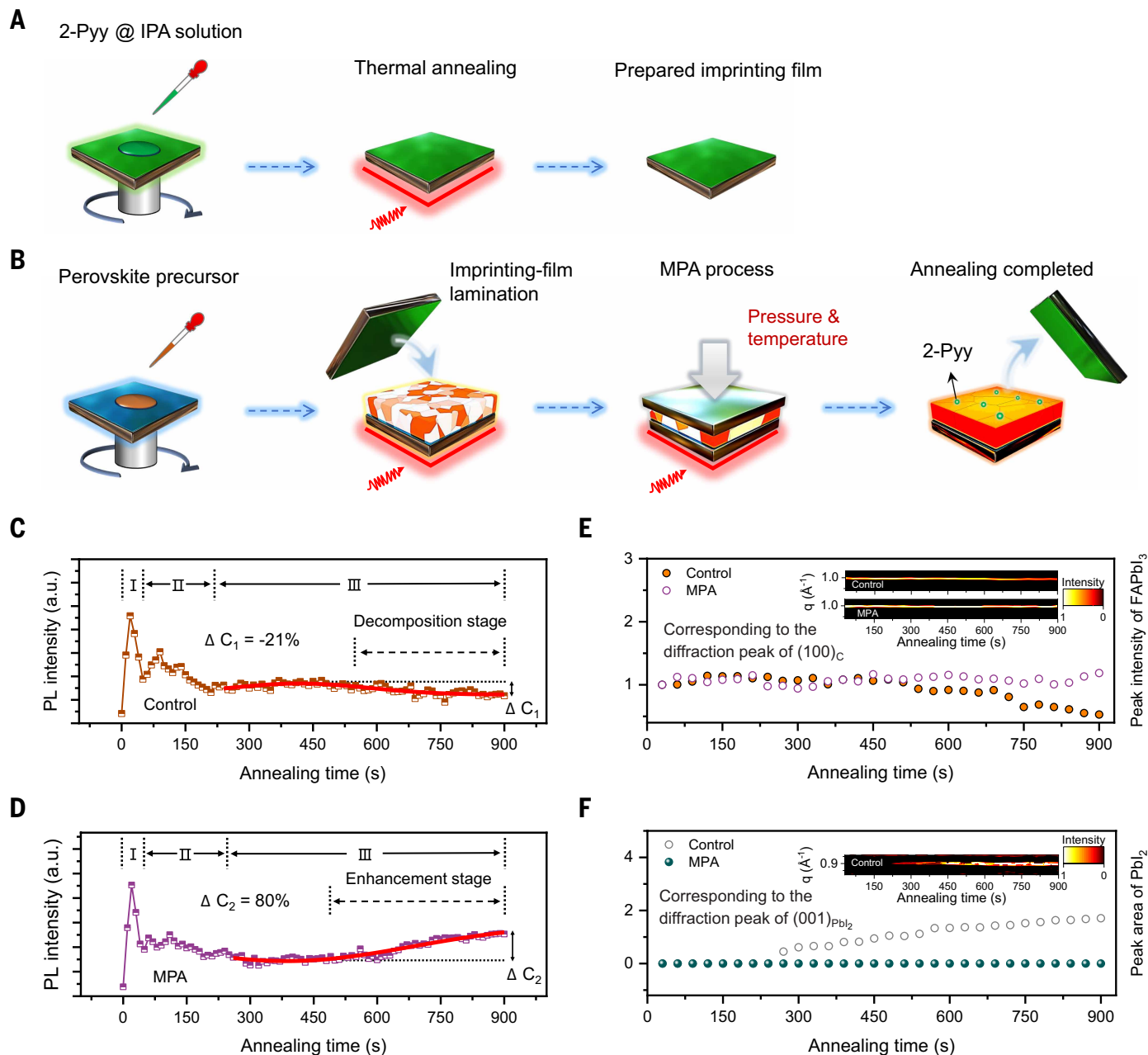


Fig. 1. MPA process and real-time characterization of perovskite film during annealing. (A) Fabrication process for the 2-Pyy imprinting film. (B) Process flow for MPA during perovskite annealing. Integrated in situ PL intensity profiles for perovskite films based on control (C) and MPA (D) as a function of annealing time. ΔC_1 and ΔC_2 represent the proportional decrease and increase in intensity, respectively, relative to the plateau. (E) Time-resolved in situ XRD intensity evolution of the FAPbI₃ perovskite diffraction peak [(100)_C, $q \approx 0.99 \text{ \AA}^{-1}$] for both control and MPA. The inset provides intensity mapping images, confirming enhanced phase stability in MPA-based films. (F) Integrated intensity of the PbI₂ diffraction peak [(001)_{PbI₂}, $q \approx 0.9 \text{ \AA}^{-1}$] over time, with the inset showing the corresponding spatial mapping for the control film, indicative of PbI₂ generation during decomposition.

as a representative perovskite and used 2-Pyy as the imprinting ligand. We first followed the surface Pb–I chemistry during annealing of the control sample using in situ x-ray photoelectron spectroscopy (XPS) at three crucial time points (300, 600, and 900 s) (fig. S5). For the control film subjected to conventional annealing, the Pb 4*f*_{7/2} peaks gradually shifted toward higher binding energies, indicating reduced electron density around Pb atoms at the perovskite surface (fig. S5A). Notably, metallic lead (Pb⁰) appeared at later annealing stages, primarily from the breakdown of Pb–I bonds in PbI₂ (19). This observation was further supported by a progressive shift of the I 3*d*_{5/2} peak from

619.22 eV (300 s) to 619.44 eV (600 s) and 619.61 eV (900 s), indicating a gradual change in the near-surface iodine chemical environment consistent with iodide loss during annealing (fig. S5B). Weakened Pb–I bonds promoted lattice distortion and allowed the escape of iodine species from the surface. Quantitative analysis of the relative atomic ratios revealed a decrease in the I/Pb ratio from 2.93 (300 s) to 2.89 (600 s) and 2.82 (900 s), corresponding to an estimated ~6.0% reduction of surface iodide (fig. S5B and table S2). However, the MPA-treated perovskite film exhibited only ~2% iodine loss after annealing (fig. S6). These data indicated progressive surface iodide loss upon prolonged

annealing, consistent with halide-deficient surfaces that expose undercoordinated lead(II) sites (note S1).

To elucidate the impact of the MPA, we used in situ optical characterization to monitor the changes in optoelectronic properties of perovskite films during annealing. Based on the variations in photoluminescence (PL) spectra (fig. S7), the entire annealing process was divided into three distinct stages (Fig. 1, C and D): In stage I, solvent evaporation induced the formation of nanoscale perovskite grains, accompanied by a marked increase in PL intensity. In stage II, grain growth and structural reorganization led to dynamic changes in the PL response (20). In stage III, the PL intensity initially plateaued around 450 s, followed by a gradual decline, indicative of annealing-induced decomposition in the control film (Fig. 1C). The gradual 21% decline in PL intensity relative to the plateau was likely associated with the accumulation of annealing-induced defects in the perovskite film. By contrast, MPA treatment yields an 80% enhancement in PL intensity, indicating a reduced defect density and improved optoelectronic quality of the perovskite (Fig. 1D).

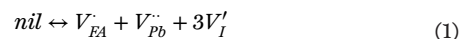
To further elucidate the effects of annealing on perovskite formation, in situ x-ray diffraction (XRD) was used to investigate the crystallization kinetics and phase evolution of perovskite materials (Fig. 1, E and F, and fig. S8). A sharp diffraction peak at $q \approx 0.99 \text{ \AA}^{-1}$ corresponding to the (100) plane of α -FAPbI₃ increased at ~ 150 s and marked the onset of crystallization driven by solvent evaporation. Notably, the (100) perovskite peak intensity gradually decreased after 450 s in the control film, suggesting thermal annealing-induced degradation of the α -phase perovskite. The absence of an α -to- δ phase transition suggested that the degradation proceeded through a nonpolymorphic pathway. By contrast, the MPA film exhibited minimal variation in peak intensity, indicating that the MPA process favored perovskite structural stability under heat treatment (Fig. 1E). To further confirm this effect, the (001) diffraction peak of PbI₂ was monitored to track its phase evolution (Fig. 1F). As annealing progressed, the trigonal PbI₂ phase ($P3m1$) appeared in the control film after ~ 300 s, indicating the onset of secondary phase formation linked to perovskite degradation. By contrast, PbI₂-related diffraction signals remained negligible in the MPA film, highlighting a strong correlation between defect accumulation and PbI₂ byproduct formation.

We used scanning electron microscopy (SEM) to examine the surface morphology of the perovskite films. At the initial stage (300 s), bright contrast regions were observed at the grain boundaries (GBs), potentially indicative of emerging secondary phases, such as PbI₂, which progressively evolved and became identifiable in later stages (fig. S9, A to C). By 600 s, distinct PbI₂ aggregates became visible at the GBs (fig. S10, A to C). However, within the first 600 s, no obvious secondary phases were observed in the surface morphology of the perovskite film based on MPA (fig. S9, D to F, and fig. S10, D to F). As annealing progressed, an increasing number of PbI₂ nanocrystals accumulated and eventually covered large areas of the control film surface (fig. S11, A to C). By contrast, only minimal PbI₂ formation was observed in the MPA samples under identical annealing condition (fig. S11, D to F). Thus, the MPA strategy effectively protected the crystals during annealing, as further confirmed by the grazing-incidence wide-angle x-ray scattering (GIWAXS) patterns, which showed no significant PbI₂ secondary phases on the MPA-treated perovskite surface (fig. S12), with $q \approx 0.9 \text{ \AA}^{-1}$.

To probe the spatial distribution of PbI₂ byproducts, grazing-incidence x-ray diffraction was conducted at variable incident angles (0.1° to 0.6°) (fig. S13). In the control sample, the PbI₂/perovskite peak intensity ratio progressively decreased with increasing incident angle, suggesting preferential PbI₂ accumulation near the film surface. Consistently, minimal PbI₂ signals were detected in the MPA sample (fig. S14). Given that defects preferentially accumulate at GBs and surfaces because of structural instability (16), these findings further support the involvement of defect-induced mechanisms in PbI₂ formation.

The working mechanism of pyridine molecules in the MPA process

First-principles calculations revealed that V_I' defects had relatively low formation energies under thermal conditions that make them thermodynamically favorable point defects in FAPbI₃ (21–23). As depicted in Eq. 1, Schottky-type defects form via the coupled loss of both cation and anion sites whereas Eq. 2 illustrates the Frenkel defect mechanism involving the displacement of iodide ions under thermal stress.



Here, *nil* denotes the ideal (defect-free) lattice, and *V* represents a vacancy. Notably, V_I' appears in both Schottky-type (Eq. 1) and Frenkel-type (Eq. 2) defects, serving as a common product for charge compensation. In addition, Eq. 3 describes the sublimation of interstitial iodine I_i during annealing, which shifts the equilibrium of Eq. 2 toward V_I' generation.

To elucidate the key function of 2-Pyy during the MPA process, we performed ab initio molecular dynamics (AIMD) simulations to investigate structural evolution under thermodynamic conditions (fig. S15). The simulations were conducted at atmospheric pressure, with the temperature from 298 to 423 K to mimic the annealing environment.

To establish a baseline, a defect-free perovskite lattice was used to model the intrinsic structural response during annealing. Throughout the annealing process, dynamic fluctuations in Pb–I bond lengths were observed, but the overall lattice framework remained intact, and no degradation products were detected (fig. S16). By contrast, in perovskite structures containing V_I' defects (Fig. 2A), the undercoordinated lead(II) cations adjacent to the V_I' sites tend to attract surrounding I[−] anions, resulting in the formation of $-PbI_x$ polyiodide complexes such as $-PbI_4$ (Fig. 2, B and C). Subsequently, the local $-PbI_4$ group migrated from its original lattice site and transformed into a free-state PbI₄ species. With continued annealing, more $-PbI_x$ groups formed around the V_I' sites, accompanied by increasing lattice distortion and ultimately resulting in Pb–I bond dissociation (Fig. 2D). We deduced that free-state PbI_x species (such as PbI₄ or PbI₃) acted as metastable intermediates that readily converted into thermodynamically stable PbI₂ under thermal annealing. The formation of V_I' defects thus served as a principal trigger for perovskite lattice collapse.

By contrast, at the onset of the simulation, 2-Pyy molecules in the MPA process rapidly established monodentate coordination with undercoordinated lead(II) sites (marked with a red circle in Fig. 2E). Notably, 2-Pyy molecules dynamically reoriented to adopt a bidentate configuration, simultaneously coordinating with two undercoordinated lead(II) centers adjacent to a single V_I' defect (marked with a red circle in Fig. 2F). This ligand-based coordination stabilized the local lattice environment, suppressing the formation of PbI_x species and mitigating the Pb–I bond dissociation typically initiated by V_I' -related structural distortions (Fig. 2, G and H).

To elucidate the ligand-based working mechanism of 2-Pyy in the MPA process, two structural analogs, 2-aminopyridine (2-Apy) and 2-picolyamine (2-Piy), were selected for comparative analysis (fig. S17). The electron-rich nitrogen atoms in the pyridine ring and aminoalkyl chain acted as dynamic anchoring sites for interaction with the perovskite lattice. Notably, variations in the length and structure of the aminoalkyl substituent modulated their inductive and conjugative interactions, thereby influencing coordination strength and molecular-level confinement.

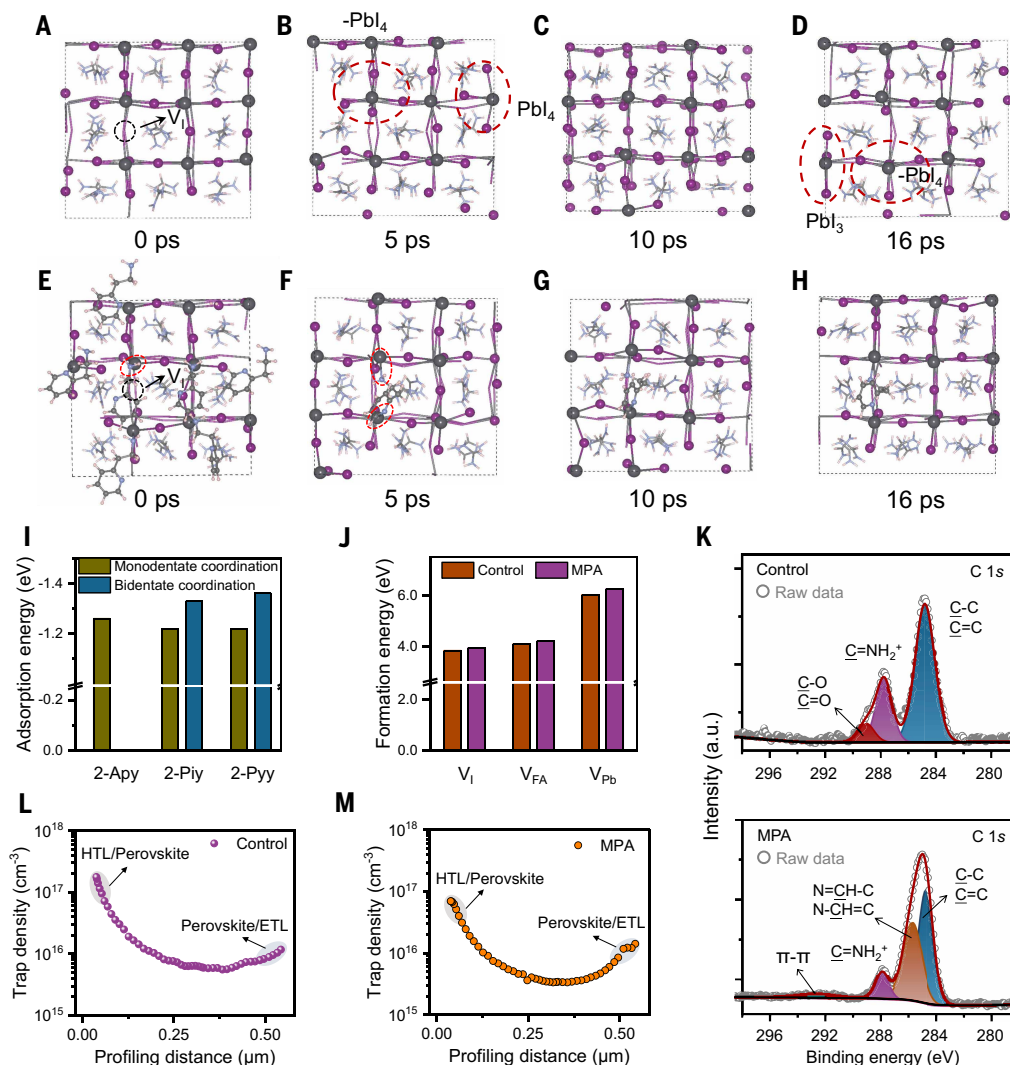


Fig. 2. Theoretical simulations and experimental validation of the functional mechanism of MPA strategy. (A to D) AIMD snapshots of perovskite supercells with V_I defects in the control film under simulated thermal annealing at 423 K. (E to H) AIMD snapshots of V_I -defective perovskite supercells in the MPA-treated film under simulated thermal annealing at 423 K, revealing enhanced structural stability and reduced dynamic disorder. (I) Adsorption energies of ligand-engineered pyridine-based molecules with distinct coordination motifs on perovskite surfaces. (J) Formation energies of vacancy defects in control and MPA films, indicating increased defect formation energy upon MPA treatment. (K) High-resolution XPS spectra of C 1s for the control and the MPA-treated films after annealing. (L and M) Depth-dependent trap density profiles measured by DLCP for the PSCs based on control (L) and MPA (M). The arrows and corresponding labels indicate specific positions in the profile. HTL, hole transport layer; ETL, electron transport layer.

Density functional theory (DFT) calculations were conducted to explore the adsorption characteristics and charge density distributions of the three pyridine-based molecules (fig. S18). Molecular size effects caused 2-Apy to preferentially adsorb at V_I defect sites with monodentate coordination through the pyridine nitrogen atom. By contrast, 2-Piy and 2-Pyy, with extended side chains, passivated the defects through bidentate coordination involving both the pyridine nitrogen and the nitrogen atom of the aminoalkyl chain, thereby exhibiting higher adsorption energies (Fig. 2I).

Fourier-transform infrared (FTIR) spectroscopy of MPA-treated films revealed a red shift in the N–H stretching vibration of the amino group was observed, from 3321 to 3305 cm⁻¹, along with a corresponding shift in the C–H stretching vibration of the pyridine ring, from 3182 to 3173 cm⁻¹ (fig. S19, A and B). These shifts revealed that hydrogen bonds formed between the 2-Pyy and the Pb–I framework. Notably, the N–H

stretching vibration of FAI also exhibited a red shift from 3359 to 3340 cm⁻¹, suggesting an additional interaction between FAI⁺ and 2-Pyy (fig. S19, C and D). Furthermore, XPS analysis revealed a shift in the Pb 4f peaks of PbI₂ toward lower binding energies, indicative of increased electron density around Pb atoms after the imprinting of 2-Pyy molecules (fig. S19, E and F). A more robust lattice structure was established.

DFT calculations were conducted to evaluate the formation energies of vacancy defects (Fig. 2J and fig. S20). With 2-Pyy adsorption, the formation energies of V_I , V_{FA} , and V_{Pb} increase from 3.86, 4.11, and 6.03 eV to 3.97, 4.23, and 6.26 eV, respectively. These increases indicated that 2-Pyy contributed to defect suppression by making vacancy formation thermodynamically less favorable, consistent with the diminished PbI₂-related emission observed in high-resolution laser scanning confocal microscopy maps (fig. S21). In situ PL mapping of half-half perovskite substrates (fig. S22) suggested that MPA effectively suppressed defect-mediated nonradiative recombination in perovskite films (fig. S23), consistent with density of states calculations which shows attenuated band-edge features and no additional gap states after 2-Pyy adsorption (fig. S24).

This observation was further corroborated by drive-level capacitance profiling (DLCP), which demonstrated a reduction in defect density at the HTL/perovskite interface induced by MPA treatment (Fig. 2, L and M, note S2, and table S3). Consistently, peaks corresponding to π – π interactions within the pyridine ring and N=CH–C/N–CH=C pyridinic bonds were observed at 292.6 and 285.7 eV, respectively, in the C 1s XPS spectra of the MPA-treated film (Fig. 2K). Notably, no detectable C=O signal (typical products of FA decomposition) was observed at ~289 eV, confirming the absence of this common perovskite degradation marker.

Film characterization

To investigate the effect of MPA treatment on perovskite film morphology, SEM characterization was performed after thermal annealing. The 2-Pyy-treated film exhibited a more uniform and compact surface compared with those treated with other imprinting ligands (fig. S25). Time-resolved terahertz spectroscopy (TR-THz) was used to investigate charge-carrier dynamics in the perovskite films (24) (Fig. 3, A and B). Because ultrafast carrier trapping occurred within femtoseconds to picoseconds (ps) after photoexcitation, the effective carrier mobility was extracted at a pump-probe delay of 5 ps under 500-nm excitation (fig. S26 and note S3). The direct current mobility of free carriers

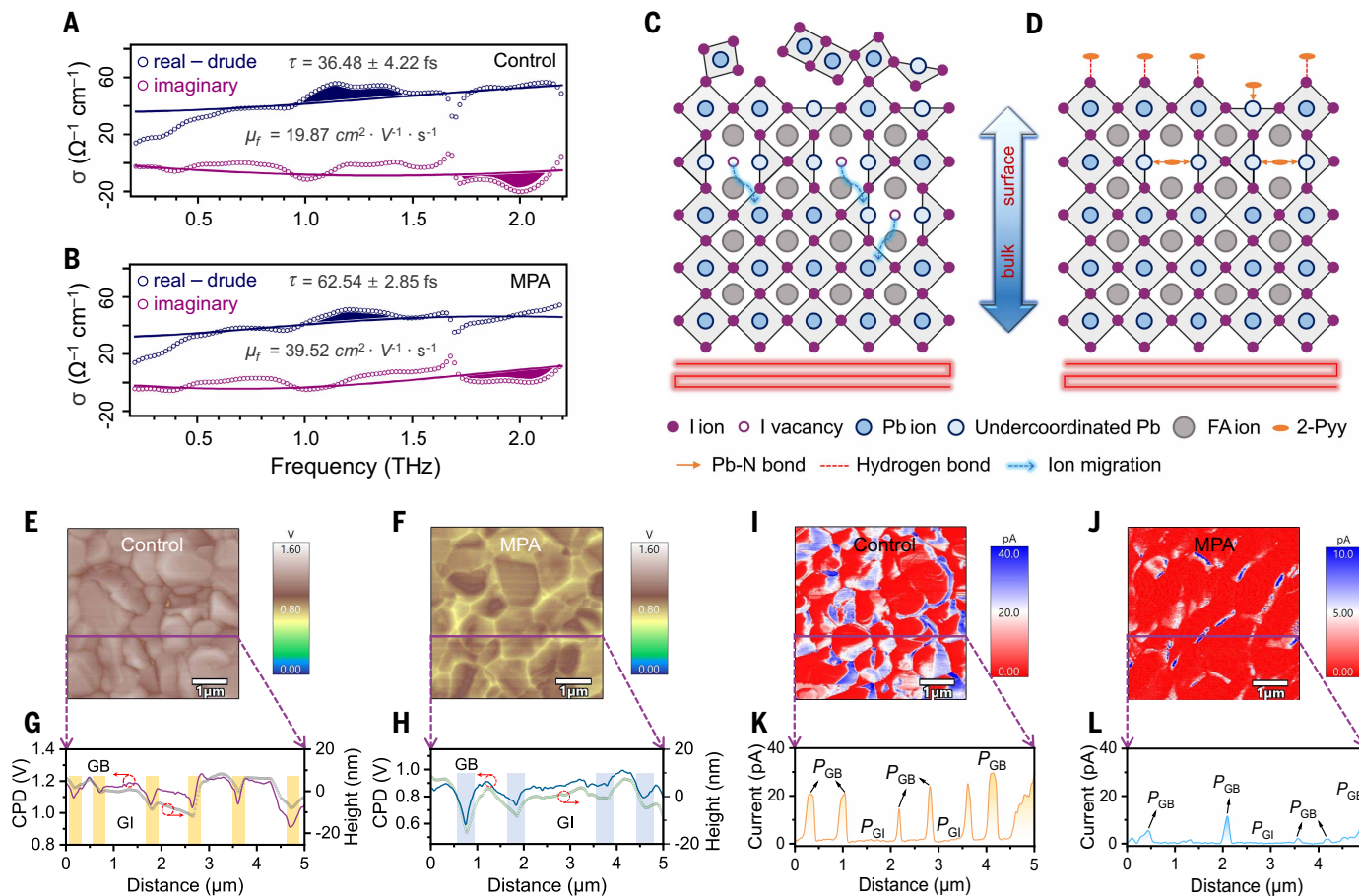


Fig. 3. Characterization of perovskite film quality and degradation mechanisms. (A and B) THz photoconductivity spectra of the control (A) and MPA-treated (B) films.

Solid circles denote the real and imaginary components of the complex photoconductivity, respectively, while solid curves represent fitted data using a Drude-Smith model. The scattering time (τ) is determined from the fitting model (solid curves). (C and D) Schematic illustrations of the perovskite degradation pathway during normal annealing (C) and the proposed MPA strategy (D). (E and F) KPFM surface potential maps of the films based on control (E) and MPA (F), reflecting enhanced surface uniformity and potential distribution upon the MPA strategy. (G and H) Line-scan profiles along the marked purple lines in (E) and (F) showing the corresponding surface topography and local CPD values for the control (G) and MPA-treated (H) films. (I and J) Conductive AFM current maps for the films based on control (I) and MPA (J), revealing more uniform and enhanced local conductivity after surface modification. (K and L) Line profiles along the purple paths in (I) and (J) displaying the lateral distribution of current flow in the control (K) and MPA-treated (L) films, P_{GB} and P_{GI} denote the positions of the GB and the GI, respectively.

in the MPA-treated film reached $39.52 \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$, versus that of the control film $19.87 \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$, as summarized in table S4. Moreover, the photoconductivity spectra revealed suppressed imaginary conductivity at ~ 1.2 and ~ 2.0 THz in the MPA film, suggesting reduced carrier localization and scattering behavior, which may be associated with trap-state mitigation facilitated by 2-Py (25). Temperature-dependent PL spectra were analyzed to extract the full-width at half-maximum, enabling quantification of the electron-phonon coupling coefficient (Γ_{op}) (note S4). The MPA-treated films exhibited a lower Γ_{op} of 13.92 meV, compared with 15.45 meV in the control, indicating weakened coupling between charge carriers and longitudinal optical phonons (fig. S27).

Figure 3, C and D, illustrates the role of MPA in modulating perovskite crystallization dynamics. In the control sample, thermal annealing induces Pb-I bond dissociation and iodine volatilization, leading to the formation of V_I defects. These vacancies accelerate perovskite degradation by promoting the formation of PbI_x intermediates and subsequent PbI_2 phase formation, while also aggravating ion migration and nonradiative recombination losses—collectively undermining device performance and structural stability. By contrast, the incorporation of MPA during annealing suppresses these failure pathways. This ligand engineering strategy leverages pyridine molecules

that anchor to the Pb-I framework, thereby increasing the formation energy of V_I defects. Their bidentate coordination arrests further defect evolution, effectively preserving the crystalline integrity of the perovskite lattice.

Kelvin probe force microscopy (KPFM) was conducted to examine the work function distribution of the perovskite films (Fig. 3, E and F). The MPA-treated film displayed a lower surface potential, indicating that the Fermi level was closer to the valence band edge (26). This enhanced p-type behavior was attributed to the suppression of n-type doping caused by V_I (10, 27). Ultraviolet photoelectron spectroscopy (UPS) further validated this shift, showing an increase in work function (WF) from 4.46 eV (control) to 4.75 eV (MPA film), along with corresponding valence band maximum (E_V) positions of -5.88 eV and -5.69 eV , respectively (fig. S28). The measured WF difference ($\Delta\phi = 0.29 \text{ eV}$) closely matched the 0.30 V surface potential shift observed via KPFM (fig. S29), confirming the consistency between the two techniques.

Line profiles of topography and local contact potential difference (CPD) are shown in Fig. 3, G and H. Notably, the MPA-treated film exhibited reduced CPD variation between GBs and intragrain (GI) regions, indicating suppressed interfacial electronic disorder likely caused by the lower defect density at GBs. Atomic force microscopy

(AFM) revealed negligible differences in surface roughness and nanocrystal size among the samples (fig. S30). To further quantify surface electronic homogeneity, two-dimensional correlation coefficients ($|R|$) between surface potential and root mean square roughness were calculated, yielding values of 0.232 for the control and 0.204 for the MPA-treated films (fig. S31 and note S5) (28). Surface potential maps of perovskite films treated with different pyridine derivatives are shown in fig. S32. The enhanced spatial uniformity of surface potential induced by MPA should facilitate more efficient carrier extraction and promote device operational stability.

Given the low migration barrier of V_p , these defects could readily accumulate or diffuse toward GBs during annealing (14, 29). To quantify this process, temperature-dependent conductivity measurements (150 K to 390 K) were conducted to extract the ion migration activation energy (E_a^{ion}) in both control and MPA-treated films (fig. S33). Notably, E_a^{ion} increased from 0.3223 eV (control) to 0.8974 eV (MPA), indicating a substantial suppression of halide ion migration after MPA treatment. Conductive atomic force microscopy was performed under dark conditions to assess the local conductivity of perovskite films (Fig. 3, I and J). The control sample exhibited pronounced current signals near GBs, indicative of localized ion accumulation (fig. S34). By contrast, the MPA-treated sample showed suppressed surface currents, consistent with enhanced defect passivation and inhibited ionic migration (Fig. 3, K and L, and fig. S35).

To further assess the role of MPA under nonstoichiometric conditions, we investigated a perovskite system containing excess PbI_2 (9%) (fig. S36). GIWAXS revealed a characteristic peak at scattering vector $q \approx 0.5 \text{ \AA}^{-1}$, attributed to the formation of PbI_2 -2-Py surface complexes. UPS and ultraviolet visible (UV-vis) absorption measurements further indicated that the complex exhibits type-I band alignment (fig. S37).

Such alignment at surfaces and GBs should alleviate charge-transport asymmetry by facilitating balanced carrier dynamics (note S6).

The effects of MPA on device performance

To evaluate the impacts of MPA strategy on device performance, we fabricated planar n-i-p structured PSCs with the configuration of ITO/ SnO_2 /FAPbI₃/spiro-OMeTAD/Au [ITO, indium tin oxide; SnO_2 , tin(IV) oxide; spiro-OMeTAD, 2,2',7,7'-Tetrakis [N, N-di(4-methoxyphenyl) amino]-9,9'-Spiro-bifluorene]. Based on the optimized conditions of 2-Py for the MPA (fig. S38), the champion MPA-treated device (0.08 cm^2) achieved a PCE of 26.6% (reverse scan) and 26.2% (forward scan) (Fig. 4A), with a stabilized PCE of 26.4% maintained over 300 s under MPP tracking (fig. S39). A well-matched current response was observed in the external quantum efficiency spectra (fig. S40). By contrast, the PSCs with conventional solution processed 2-Py passivation served as the control achieved PCEs of 25.2% (reverse) and 24.5% (forward). Statistical photovoltaic parameters indicated that MPA-based devices outperform those using conventional methods, primarily through concurrent improvements in open-circuit voltage (V_{OC}) and fill factor (FF) (fig. S41). Figure S42 shows the champion PCEs of devices based on 2-APy, 2-Py, and 2-Py molecules, with the MPA strategy effectively mitigating hysteresis. A certified PCE of 26.5% was achieved using 2-Py, with a V_{OC} of 1.18 V, a J_{SC} of 26.4 mA cm^{-2} , and an FF of 84.6% under reverse scan (fig. S43). The MPA approach scaled to 1 cm^2 PSCs, which delivered PCEs of 24.9% (reverse scan) and 24.0% (forward scan) (Fig. 4B). To further assess the scalability of the MPA strategy, we fabricated perovskite solar modules with an active area of $\sim 16.0 \text{ cm}^2$ (fig. S44). The champion module yielded a PCE of 23.0% (reverse scan) and 22.5% (forward scan) and maintained a stabilized PCE of 22.5% under MPP tracking for 300 s, with good reproducibility.

We used thermal admittance spectroscopy to probe both shallow and deep-level defect states in the devices (fig. S45 and note S7). The MPA-based PSCs exhibited a reduced trap density of states relative to the control, correlating with the observed performance enhancement (30). Additionally, space-charge-limited current (SCLC) measurements performed under dark conditions (fig. S46) revealed a lower trap density in 2-Py-based devices ($9.97 \times 10^{15} \text{ cm}^{-3}$) compared with the control ($1.62 \times 10^{16} \text{ cm}^{-3}$), 2-APy ($1.39 \times 10^{16} \text{ cm}^{-3}$), and 2-Py ($1.25 \times 10^{16} \text{ cm}^{-3}$). Furthermore, impedance-frequency analysis (fig. S47) showed markedly suppressed low-frequency responses in MPA-treated devices, with the extracted ionic conductivity decreasing from $3.0 \times 10^{-7} \text{ S cm}^{-1}$ in the control to $2.4 \times 10^{-7} \text{ S cm}^{-1}$ after MPA treatment, indicating effective mitigation of defect-mediated ion migration (31).

The application of the MPA strategy enhances the stability of perovskite films and devices under illumination and environmental exposure. Iodine-related species, particularly volatile I_2 , can trigger chemical degradation and accelerate perovskite decomposition through redox reactions (5, 29, 32). To evaluate the role of MPA in mitigating iodine-induced degradation, we examined the reactivity of three pyridine-based molecules with I_2 . Among them, 2-Py exhibited the fastest I_2 reduction rate (fig. S48), indicating its superior potential in reducing iodine species. UV-vis characterization confirmed the effective reduction of I_2 by 2-Py (fig. S49).

To further evaluate the role of MPA in mitigating iodine-induced degradation, we conducted accelerated aging tests by exposing the films to iodine vapor under illumination (fig. S50). The MPA-treated film retained both its visual appearance and photoluminescence

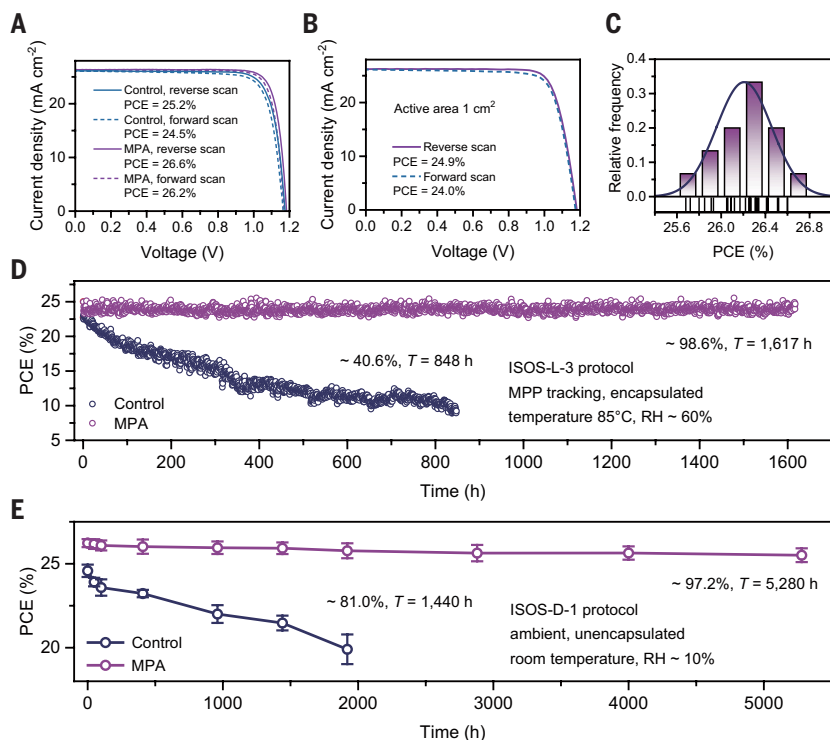


Fig. 4. Photovoltaic performance and universality of the MPA strategy. (A) The J - V curves of PSCs fabricated with traditional solution-processed passivation (control) and the solid-state MPA strategy. (B) The J - V curves of a larger device (1 cm^2) based on the MPA strategy. (C) Statistical analysis of devices performance for testing the reusability of the MPA technique. (D) Operational stability of MPP and MPA devices. Encapsulated devices were evaluated under continuous LED illumination at the MPP under RH of $\sim 60\%$ and a temperature of 85°C , following the ISOS-L-3 protocol. T refers to the aging time of different devices. (E) Long-term storage stability of unencapsulated control and MPA-treated devices under ambient conditions (room temperature, RH $\sim 10\%$) in the dark, following the ISOS-D-1 protocol.

intensity whereas the control film exhibited pronounced degradation. We used FTIR to map the chemical stability of perovskite films after prolonged light exposure (fig. S51). The absorption band at 1710 cm^{-1} , corresponding to the symmetric bending mode of $\text{C}=\text{NH}_2^+$ in FA^+ , exhibited much smaller variation in MPA films than in the control after 300 hours, further confirming the stabilizing effect of the MPA strategy.

Moreover, the MPA strategy markedly enhances the surface hydrophobicity of perovskite films (fig. S52). The increased hydrophobicity may stem from ligand-induced surface modification, which passivates polar defects and reduces water affinity (33, 34). Consistently, PL mapping revealed that MPA-treated films retained stronger luminescence under humid conditions, evidencing enhanced moisture resistance (fig. S53). Furthermore, thermal stability was enhanced in MPA-treated films relative to the control samples (fig. S54).

Device operational stability was monitored at the MPP condition following the ISOS-L-3 protocol (35) (Fig. 4D). To mitigate degradation associated with the HTLs, we replaced spiro-OMeTAD with poly(triaryl amine) and conducted accelerated damp heat testing at 85°C and $\sim 60\%$ RH. Both the MPA-based and control devices were encapsulated, with the former retaining $\sim 98.6\%$ of its initial PCE (25.0%, fig. S55) after 1617 hours, whereas the latter exhibited a 40.6% reduction of its initial efficiency (22.7%, fig. S55) after 848 hours. In comparison with high-performance PSCs evaluated under the ISOS-L-3 protocol (table S5), our devices demonstrate highly competitive operational stability. Long-term storage stability of unencapsulated devices was further evaluated under ambient conditions (room temperature, $\sim 10\%$ RH), in accordance with the ISOS-D-1 protocol. After 5280 hours, MPA-treated devices retained $\sim 97.2\%$ of their initial PCE, in contrast to $\sim 81.0\%$ retention for the control devices after 1440 hours (Fig. 4E). For the devices based on spiro-OMeTAD (fig. S56), the MPA device retained 91.6% of its initial PCE after 500 hours under ISOS-L-3 conditions, compared with 66.9% after 170 hours for the control devices.

To evaluate the sustainability of the MPA strategy, a single 2-Pyy imprinting film was reused for fabricating 30 consecutive PSCs. High device performance repeatability (Fig. 4C and fig. S57) and retention of film properties after reuse (figs. S58 and S59) further substantiate the high reliability of the MPA strategy. During batch device fabrication, MPA reduced overall processing costs by more than 47 times compared with conventional solution-processed passivation, owing to its solvent-free nature and minimal material waste (table S6). Importantly, the MPA approach is compatible with a wide range of perovskite compositions, including MAPbI_3 , $\text{Cs}_2\text{FA}_{1-x}\text{PbI}_3$, and $\text{MA}_x\text{FA}_{1-x}\text{Pb}(\text{I}_y\text{Br}_{1-y})_3$ (figs. S60 to S62), highlighting its broad applicability. The MPA strategy is also applicable to inverted (p-i-n) PSCs, yielding a notable efficiency enhancement and confirming its compatibility with various device architectures (fig. S63).

Discussion

This work establishes that iodide loss during thermal annealing is a key driver of structural degradation in polycrystalline perovskite films. We introduced a MPA process designed to suppress defect initiation from iodide loss and preserve lattice integrity at the source. The selected ligand molecules (2-Pyy) in MPA form a solid-state bidentate coordination complex with the undercoordinated lead(II) sites on the perovskite surface, thereby effectively mitigating further defect propagation and enhancing the device performance. MPA demonstrates strong scalability across diverse perovskite systems and high compatibility with fabrication of the large-area modules. This study elucidates the degradation kinetics of perovskite under practical processing and presents an effective remedy poised for deployment in industrial production of perovskite photovoltaics.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S63; Tables S1 to S6; References (36–70); Movie S1

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